
HaluRISC: Calibrated, Explainable Hallucination Risk Prediction for LLM Outputs

Submitted By

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1 Project Overview

Large language models (LLMs) like ChatGPT are very good at writing text that sounds confident and correct, but they sometimes generate information that is wrong or made up — this is called a **hallucination**. This project builds a complete system that predicts **how likely an LLM answer is to be hallucinated**. The system is **black-box friendly** — it only needs the question, the answer, and (optionally) some context, without ever looking inside the LLM. It is **lightweight** — it uses simple text features and classical machine learning (XGBoost), so it runs in milliseconds on a normal computer, with no GPU and no API fees. It is **explainable** — for every prediction it shows *why* the answer looks risky (for example: “the answer contradicts the provided context”, or “the answer contains numbers that are not supported”). Finally, it is **presentable** — a polished, interactive web dashboard lets a judge or supervisor test the system live, type any question and answer, and see the risk score and explanation instantly. The project is judged on two tracks: the quality of the machine learning, and the quality of the final product — the dashboard is designed from day one to be the centerpiece of the demo.

2 Motivation

LLMs are now used in real products — chatbots, Q&A systems, content review tools — where a wrong answer has a real cost, and the most common way to detect hallucinations is to ask the LLM itself to judge its own output, which is slow, expensive, and sometimes wrong. A lightweight model trained on simple text features (word overlap, named entities, contradiction with context, numbers, hedging words) can flag risky answers almost instantly and at almost zero cost. A risk score is also more useful than a simple “yes/no” flag, because real systems can decide what to do with a risky answer — show a warning, ask for human review, or fetch better evidence. And because the final result is a working product with a clean UI rather than just a report, the project is fully demo-able and easy to present.

3 Objectives

Research question: Can a lightweight machine learning model predict hallucination risk in LLM answers using simple text features, while giving calibrated risk scores and clear explanations? To answer this, the project has five objectives: (1) build a feature extraction pipeline (about 20–30 features) that captures how well an answer agrees with the question and context — word overlap, named entities, contradiction signals, numbers, hedging words, and semantic similarity; (2) train and honestly compare classical models — a simple heuristic baseline, Logistic Regression, Random Forest, and XGBoost; (3) calibrate the risk scores so that a score of 0.8 truly means “about 80% risk”, making the score trustworthy for real use; (4) explain predictions with SHAP, giving a global view of which features matter most plus detailed case studies of individual predictions; and (5) build a **polished web dashboard** (React + FastAPI) with strong UI/UX — live prediction, risk gauge, explanation panel, and a demo gallery — designed for presentation and judging.

4 Methodology

The pipeline has seven clear steps:

1. **Data preparation** — load the HaluEval QA dataset, clean the text, and manually inspect 50 random samples to verify label quality before training.
2. **Feature extraction** — compute features in seven groups: length/style, lexical overlap, entity overlap, NLI contradiction (using a lightweight pre-trained NLI model), numeric consistency, hedging phrases, and semantic similarity.
3. **Model training** — 5-fold stratified cross-validation, hyperparameter tuning for XGBoost, and every experiment repeated with 3 random seeds, reported as mean $\pm$ standard deviation.
4. **Model comparison** — heuristic baseline, Logistic Regression, Random Forest, and XGBoost, judged on F1, AUROC, precision, recall, and MCC.
5. **Calibration** — apply Platt scaling to the final model and measure the gain with Expected Calibration Error (ECE) and Brier score.

6. **Explanation** — SHAP global feature importance plus 3 local case studies; error analysis on 20 wrong predictions (10 false positives, 10 false negatives).
7. **Deployment** — FastAPI backend serving the trained model, and a React (Vite) dashboard with a clean, modern UI: risk gauge, feature breakdown, and live input panel.

Statistical rigor: model differences are tested with McNemar’s test and bootstrap confidence intervals, so conclusions are not based on luck of the split.

External comparison: results are compared with published HaluEval detection baselines (e.g., SelfCheck-GPT) so a judge can see whether the scores are good or bad.

5 Dataset

We use the **HaluEval** benchmark (EMNLP 2023) — the standard dataset for hallucination evaluation, widely used and well documented.

- **Size:** 35,000 samples across 4 task types (QA, dialogue, summarization, general).
- **Used here:** the **QA subset** (about 10,000 samples). Each sample has a question, an answer, a context, and a label (hallucinated or not).
- **Why QA:** it is the largest, cleanest, and most comparable subset, and it matches the demo scenario (question → answer → risk score).

6 Expected Outcomes

1. A journal-style project report (methodology, experiments, results, error analysis, limitations).
2. A trained, calibrated XGBoost model with feature ablation and SHAP explanations.
3. Statistical significance tests (McNemar, bootstrap confidence intervals).
4. **A polished, judge-ready web dashboard** — live risk prediction with clean UI/UX.
5. Reproducible code: dataset preparation, feature extraction, training, and API.

7 Work Plan (8 Weeks)

Week	Focus	Main Deliverable
1	Literature + dataset setup	HaluEval QA loaded, 50 samples manually audited
2	Core features	Length, lexical overlap, hedging, numeric features + NLI model setup
3	Advanced features + baselines	Entity, semantic, NLI features; Logistic Regression and Random Forest results
4	XGBoost + calibration	Tuned XGBoost, 3-seed results, Platt calibration, ablations
5	Explanations + statistics	SHAP analysis, error analysis, McNemar + bootstrap tests
6	Backend API	FastAPI service with model + calibration + SHAP
7	Frontend dashboard	React dashboard, UI/UX polish, demo gallery
8	Report + demo	Final report, demo rehearsal, submission

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